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MUSCULOSKELETAL PAIN

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

CBT Cognitive Behavioural Therapy

IM Intramuscular

LBP Low Back Pain

OTC Over-the-Counter

PDE5 Phosphodiesterase Type 5

PT Physiotherapy

SLR Straight Leg Raise

TENS Transcutaneous Electrical Nerve Stimulation

OBJECTIVES

For each topic (low back pain, neck pain and whiplash, and sports injuries):

- 1 Understand the pathophysiology, clinical presentation, and diagnosis
- 2 Identify risk factors
- 3 Determine the goals of therapy
- 4 Learn to treat with both non-pharmacological measures and pharmacological measures
- 5 Understand how to monitor for relief

LOW BACK PAIN

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PATHOPHYSIOLOGY

Low back pain: pain ranging from mild to severe originating from the lumbosacral region (may extend to the gluteal muscles, hips, and low extremities)

The types of back pain are based on how long the patient has been experiencing the pain



Recurrent back pain: back pain that subsides then reoccurs less than 6 months after the initial episode

CAUSES OF LOW BACK PAIN

- Most low back pain stems from mechanical causes
- Non-mechanical causes are when the pain is caused by something not activity-related
 - Osteoporosis causing degeneration of vertebrae
 - Arthritis affecting the spine

Mechanical
Muscle or ligament strains
Herniated or ruptured disks
Traumatic injuries

Non-Mechanical
Arthritis
Skeletal stenosis
Sciatica
Osteoporosis

CLINICAL PRESENTATION AND DIAGNOSIS

Clinical Presentation

- Muscle aches
- Pain radiating to the leg
- Difficulties standing up straight
- Reduced flexibility and range of motion of the back
- Radicular low back- paresthesia with or without weakness

Diagnosis

FYI

- Physical exam done in primary care
- Most cases do not require diagnostic imaging
 - May refer for further workup if any red flags present

RISK FACTORS – LOW BACK PAIN

- Age >30 years old
- Sedentary lifestyle
- Weakened abdominal muscles
- Conditions affecting the alignment of the spine (e.g. scoliosis)
- Obesity
- Occupations requiring heavy lifting or bending
- Comorbidities affecting the back or spine (cancer, osteoarthritis, etc.)
- Mental health conditions may cause low back pain (depression, anxiety)
- Pregnancy

RED FLAGS

Condition	Red flags in patients experiencing low back pain
Cancer	Previous cancer history, unexplained weight loss, age > 50, failure to improve after 1 month of treatment
Cauda equina syndrome (Medical emergency)	Loss of anal sphincter tone, fecal incontinence, perineal numbness, weakness of legs, change in sexual function, acute urinary retention or overflow incontinence
Infections e.g. • Epidural abscess • Spinal osteomyelitis	History of intravenous drug use or other sources of infection, paraparesis, quadriparesis, bowel/ bladder impairment, fever, sensory loss, local or radicular pain unrelieved by position change History of intravenous drug use or other sources of infection, fever
Spinal fracture or compression fracture	Age >50, history of osteoporosis or use of corticosteroids, major trauma
Herniated nucleus pulposus	Sharp or burning pain, numbness or tingling, muscular weakness, radiating pain, pins and needles



GOALS OF THERAPY

- ✓ Acute low back pain
 - ✓ Help patients recover quickly to prevent progression to chronic pain
 - ✓ Reduce pain and increase activity
- ✓ Subacute low back pain
 - ✓ Help patients recover and prevent progression to chronic pain
 - ✓ Maintain functional goals
- ✓ Chronic low back pain
 - ✓ Help patients return to their normal daily activities
 - ✓ Manage psychological symptoms
 - ✓ Maintain functional goals-self management

Overall: reduce pain and increase function!

NON-PHARMACOLOGICAL MEASURES

Acute Low Back Pain

- Encourage continuation of normal daily activities
- Avoidance of bed rest
- Apply superficial heat/cold for short-term relief (max 15-20 minutes at a time, not directly on the skin)

Subacute/Recurrent Low Back Pain

- Patient education
- Planned work re-entry
- Active exercising
- Avoidance of bed rest
- Coping skills
- Avoiding strain
- May combine the above with the following:
 - Ice/heat
 - Laser
 - TENS
- Combining treatment modalities with exercise has increased benefits

Chronic Low Back Pain

- May be beneficial in patients who practice >2 of the following:
- Physical activity
 - CBT or stress reduction program
 - Self-management programs
 - Coping skills
 - Increase functioning
 - Increase confidence to manage symptoms
 - Yoga, relaxation techniques
 - Physical conditioning programs (prescribed walking program)
 - Bio-psychosocial model of care: interdisciplinary
 - Massage therapy (adjunct)

YELLOW FLAGS

Yellow flags are psychosocial behaviours that prolong recovery and increase the risk of prolonged disability and work loss



Yellow Flag	Intervention
Pain and activity are harmful	Patient education, CBT
Extended rest	Patient education, rehab
Social Withdrawal	Referral for mental health care
Problems with work	Refer to a case manager
Lack of support	Patient education, family educate
Issues with claims and compensation	Refer to a case manager

PHARMACOLOGICAL MEASURES

ACUTE AND SUBACUTE LOW BACK PAIN

NSAIDs (e.g. ibuprofen, naproxen, diclofenac, celecoxib, ketoprofen)

- **No specific NSAID is recommended** but shown to help with pain relief
- Consider side effect profiles and patient-specific factors (e.g. celecoxib has the lowest GI risk)

Acetaminophen

- Better safety profile than NSAIDs but **considered ineffective**
- Reasonable option if other medications not tolerated

Muscle relaxants (e.g. cyclobenzaprine, baclofen, tizanidine, benzodiazepines, preparations containing methocarbamol or orphenadrine)

- May temporarily relieve pain and spasms with benefits seen within one week
- Used short-term (for about 5-7 days)
- May cause drowsiness and dizziness

Short-acting opioids (acetaminophen + codeine or tramadol)

- Reserved for severe pain, not commonly used

PHARMACOLOGICAL MEASURES

CHRONIC LOW BACK PAIN

Acetaminophen

- **Generally ineffective** but good safety profile
- May be an option depending on patient factors

NSAIDs

- Some evidence to support the use of NSAIDs
- Consider side effects and patient factors

Muscle relaxants

- Not as effective for chronic pain, can be used short-term for flares of pain

GABA derivatives (e.g. gabapentin, pregabalin)

- **May be useful if pain is in multiple areas**

Antidepressants

- **Most antidepressants have no evidence they relieve chronic low back pain**; however, they may help with depression or anxiety associated with the condition
- Bupropion, SSRIs, and TCAs may be used for comorbid depression in chronic low back pain
- **Duloxetine may provide a small benefit and has a Health Canada approved indication for chronic low back pain**
 - Good option for elderly patients due to GI and CV risks associated with chronic NSAID use



Overall lack of high-quality evidence

PHARMACOLOGICAL MEASURES

CHRONIC LOW BACK PAIN

Opioids

- **Avoid in most patients** as harms likely outweigh benefits
- Indicated for **moderate-severe symptoms**
- Consider if pain is persistent despite the use of non-opioid analgesics
- Tramadol (+/- acetaminophen) has limited evidence of being more effective than placebo for uncomplicated cases
- **Try weaker opioids first** (tramadol or codeine + acetaminophen) before stronger controlled-release opioids (morphine, hydromorphone, oxycodone)
- Consider long-acting opioids when patients are using short-acting agents frequently
- Buprenorphine patch may be useful if significant central sensitization and myofascial pain
 - Well tolerated in elderly
 - Useful in renal impairment
 - ~20% of patients develop contact dermatitis from patch adhesive

MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Low back pain	Reduce or resolve	Ongoing
Progression of pain to chronicity	Avoid	
Functional goals, activity	Maintain or increase	
Drug therapy adverse effects	Decrease or eliminate	

THERAPEUTIC TIPS

- A small number of patients with chronic low back pain may have a comorbid mental health condition (such as depression or anxiety) that could be masked by the presentation of pain – **always assess for comorbid mood disorders**
- Patients who have trouble sleeping should consider using a caffeine-free analgesic
- Consider referral to a pain management clinic, especially in cases of severe, treatment-resistant chronic back pain

CASE 1

RJ is a 24-year-old male who presents to your pharmacy with low back pain. RJ describes his pain as a "dull throbbing pain" and states it has been present for a few days. RJ recently started playing golf and he is adapting to the new sport well; he was able to get his first hole-in-one over the weekend! He has no medical conditions however he does take a multivitamin daily. RJ does not drink alcohol or consume any caffeine. He has tried using topical diclofenac for his lower back but states it isn't providing him with enough relief to continue with his daily activities. He states he gets hives when taking penicillin. He asks you for a recommendation so he can get back to practicing his swing.

1. What is the most appropriate recommendation for RJ?
 - a) Tramadol
 - b) Acetaminophen
 - c) Pregabalin
 - d) Naproxen



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2. When would you reassess RJ's pain to determine if his new medication is working for him?
- a) 2-3 days
 - b) 3-5 days
 - c) 5-7 days
 - d) 7-10 days



NECK PAIN AND WHIPLASH

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PATHOPHYSIOLOGY

- NECK PAIN
 - Condition commonly seen in primary practice
 - **Key symptom of whiplash**
- WHIPLASH
 - Hyperextension-hyperflexion mechanism transferring energy to the neck area, that can result in bone or soft-tissue injuries
 - Can occur from a motor vehicle accident, a sports accident, physical abuse, or other trauma

Acute Phase

- **Days 1-7**
- **Weeks 1-4**



Subacute Phase

- **4-12 weeks**



Chronic Neck Pain

- **≥ 6 Months**

CLINICAL PRESENTATION AND DIAGNOSIS

Clinical Presentation

- Neck pain or stiffness that gets worse with movement
- Reduced range of motion of the neck
- Headache
- Tenderness or pain in the shoulder or upper back
- Fatigue
- Dizziness

Diagnosis

FYI

- Acute Phase 1 to 7 Days
 - Patient history/physical examination
 - Decide whether X-rays are required based on neurological symptoms, impaired consciousness, and trauma history (suspicion of a cervical fracture)

YELLOW FLAGS

Yellow flags are psychosocial behaviours that prolong recovery and increase the risk of prolonged disability and work loss



Yellow Flag

Unexplained disability and high-intensity pain that is self-reported

Belief activity is harmful

Mental health conditions

Previous MSK symptoms

Passive coping skills

Avoiding movement

Increased risk of progressing to chronic neck pain

Reports severe pain

Pain in multiple areas

Seeks medical attention early on

Females

Older patients

Previous neck pain or whiplash

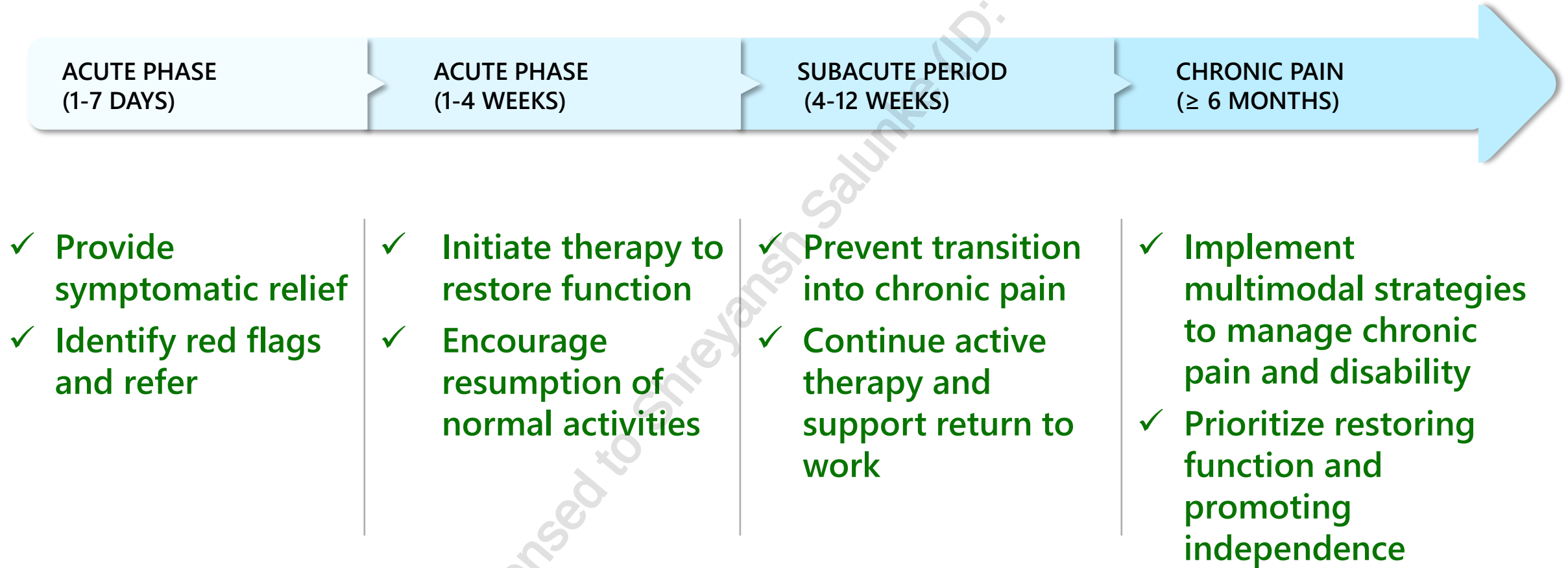
Pervasive functional impairment

RED FLAGS

Condition	Red flags in patients experiencing neck pain
Neurologic Pathology	Compression of the cervical spine, demyelination, signs of progressive loss of neurologic status
Infection	Fever, stiff neck, altered mental status , intravenous drug use, other sources of infection, patients on immunosuppressive therapies
Fracture	Trauma leading to neck pain, osteoporosis
Tumor	Unexplained weight loss, significant night pain, severe fatigue, history of cancer
Inflammatory	History or symptoms of Giant-cell arteritis, polymyalgia rheumatica, rheumatoid arthritis



GOALS OF THERAPY



CLASSIFICATION

FYI

Whiplash-Associated Disorders (WAD) Criteria

WAD-I	Neck pain, stiffness, tenderness, no physical signs or limitations
WAD-II	Pain, tenderness, reduced range
WAD-III	Pain, neurologic signs, sensory or motor reflex changes without fractures or instabilities
WAD-IV	Fracture or dislocation

Bone and Joint Decade 2000-2010 Classification

GRADE I	Neck pain/associated disorders without indication of major structural pathology, and minor/no interference with daily activities
GRADE II	No indication of major pathology; major interference with daily activities
GRADE III	No indication of major pathology; has neurologic signs (abnormal reflexes, weakness, sensory deficits)
GRADE IV	Signs/symptoms of major structural pathology (may include fracture, vertebral dislocation, spinal cord injury, infection, neoplasm, systemic inflammatory disease)

NON-PHARMACOLOGICAL MEASURES

WAD I TO WAD III – Acute Phase (Weeks 1-4)

- Symptom relief as needed
- Resume normal activities as soon as possible
- Patient education on condition
- Counseling on posture or sleep modifications if needed
- Advise to avoid unnecessary immobilization or complete bed rest
- Limit absence from work



WAD IV requires an urgent surgery consultation and management

NON-PHARMACOLOGICAL MEASURES

SUBACUTE PHASE (4-12 WEEKS)

Based on available evidence	• Exercise has more evidence than passive therapies, but combining both may help patients with symptoms • Psychoeducation and active PT reduce disability during this period
Limited evidence	• Taking breaks during sedentary work
Poor quality evidence	• Massage
Moderate quality evidence	• Improved functioning and moderate pain relief with stretching and strengthening exercises
Limited low-quality evidence	• Manual therapies (e.g. mobilization/manipulation of the spine)

NON-PHARMACOLOGICAL MEASURES

CHRONIC NECK PAIN (≥6 MONTHS)

- Improve function and well-being
 - Patients may develop mental health conditions due to chronic pain over time → **always assess patients for symptoms of depression and anxiety**

Expert opinion	• Multidisciplinary pain management is recommended in patients with chronic pain, impaired functioning and inability to return to work • Individual supervised active exercise
Limited evidence	• Multimodal/multidisciplinary treatments (e.g. CBT, exercise and psychosocial interventions, programs helping patients plan for work re-entry)
Weak and limited evidence	• Acupuncture, traction therapy, dry needling, TENS

PHARMACOLOGICAL MEASURES

WAD I TO III- ACUTE PHASE 1 TO 4 WEEKS

NSAIDs

- **More effective in relieving symptoms than placebo during the first month**
- No specific NSAID is recommended, so use side effect profiles and patient-specific factors to choose the best option

Alternative to NSAIDs

- Tramadol, codeine (if other treatments ineffective)
- **Muscle relaxants** (e.g. cyclobenzaprine, baclofen, tizanidine, orphenadrine)
- IV methylprednisolone: not recommended until more robust studies available
- Acetaminophen: evidence shows not effective in short term

PHARMACOLOGICAL MEASURES

SUBACUTE PHASE (4-12 WEEKS)

Expert opinion to alleviate pain or tenderness in more than one area	• Duloxetine • TCAs • GABA derivatives • Tramadol • Muscle relaxants
Limited evidence	• Muscle relaxants - expert opinion still recommend their use
Some available evidence	• Intramuscular lidocaine + stretching exercises
Weak and contradictory evidence	• NSAIDs

If a patient is experiencing severe pain, consider opioids

PHARMACOLOGICAL MEASURES

CHRONIC NECK PAIN (≥ 6 MONTHS)

Expert opinion	• Muscle relaxants • GABA derivatives • Tramadol
Weak and contradictory evidence	• NSAIDs
Limited evidence	• Opioids
Limited and poor-quality evidence	• Epidural corticosteroids • Lidocaine
Not effective	• TCAs • Botox



The medication selected should be based on patient-specific factors as evidence is limited

MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Patient comfort	Improvement	Days
Patient returns to work/normal daily functioning	Return to baseline	Individual timeframe As soon as able

THERAPEUTIC TIPS

- Refer all red flags
- Focus on goals of therapy – prevent long-term impairments and the progression to chronic pain while helping the patient return to their normal activities as soon as possible
- If initiating pharmacological measures also always include non-pharmacological measures
- Acute neck pain: non-pharmacological measures are usually sufficient for symptom management; if not sufficient, consider NSAIDs
- Subacute neck pain: consider muscle relaxants, duloxetine, TCAs, GABA derivatives, or tramadol
- Chronic neck pain: consider muscle relaxants, GABA derivatives, and tramadol
- Opioids may be considered when pain is severe and nonresponsive, and the 2017 Canadian Guideline for Opioid Therapy and Chronic Non-Cancer Pain should be considered
- Evidence for neck pain treatment is somewhat limited, always consider individual patient factors when selecting pharmacologic and non-pharmacologic measures

CASE 2

PO is a 34-year-old female who was involved in a rear-end motor vehicle collision. She is experiencing nausea, headache, neck pain, and limited mobility of her neck. Her CT scan and X-ray ruled out any serious injuries or other illnesses. She describes her pain as achy and rates it 4/10. PO is looking for a pharmacologic recommendation to help with her pain; however, she is open to discussion as she does not want to harm her baby since she is 3 months pregnant. She has private insurance coverage through work and would also appreciate any referrals you think are appropriate.

1. What is the most appropriate recommendation for PO?
 - a) Ibuprofen
 - b) Acetaminophen
 - c) Topical diclofenac 2.2%
 - d) Referral to a physiotherapist



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2. Your recommendation in #1 was not sufficient for pain relief, and PO returns to you for more help. Which option would be most appropriate for PO to try?

- a) Codeine
- b) Tramadol
- c) Cyclobenzaprine
- d) Methylprednisolone



SPORTS INJURIES

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PATHOPHYSIOLOGY

Examples of common sports injuries:

- Supraspinatus Tendinosis (Rotator Cuff Disorders)
- Lateral Epicondylitis ("Tennis Elbow")
- Medial Epicondylitis ("Golfer's Elbow")
- Patellofemoral Pain Syndrome ("Runner's Knee", chondromalacia patellae)
- Lower Leg Pain ("Shin Splints")
- Stress Fractures of the Tibia or Fibula
- Ankle Sprain
- Achilles Tendinitis
- Plantar Fasciitis



Ligaments bind bone to bone, tendons bind muscle to bone

CLINICAL PRESENTATION

- Pain
- Inflammation
- Muscle weaknesses
- Reduced flexibility
- Reduced weight-bearing capabilities
- Reduced range of motion

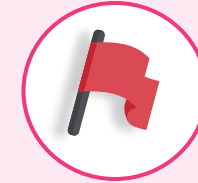
DIAGNOSIS

FYI

- **Muscle spasm:** involuntary muscle contraction
- **Muscle cramp:** prolonged muscle spasm that is painful
- **Bursitis:** inflammation of the bursa (fluid-filled sacs located between bones and muscles or tendons near joints) due to repetitive motion or acute injury in the elbow, knee, or shoulder
- **Sprain:** ligament injury due to overstretching or twisting that can cause pain, swelling, and tenderness with bruising around the injury. Sometimes can be hard to distinguish from a fracture, so an X-ray is used to diagnose
- **Strain:** muscle injury caused by overstretching that causes pain and swelling
- **Plantar fasciitis:** inflammation of plantar fascia resulting in heel pain due to increased and/or repetitive activity such as jogging
- **Shin splints:** excess strain is put on muscle attached to shins causing tearing and inflammation around the tibia (shin) resulting in pain along the inner edge of the shins, commonly occurs in running
- **Stress fractures:** mini bone fractures caused by repetitive impact; commonly in lower extremities (e.g., ankles, feet, legs); pain decreases with rest and increases over time, gets worse when pressure is applied during activity
- **Tendinitis:** inflammation of a tendon due to overuse or injury resulting in pain and tenderness just above the heel (Achilles tendinitis); or pain and tenderness at or below the elbow joint (tennis elbow)

RED FLAGS

- More than 2 weeks since injury occurred
- Age: < 2 years or > 50 years
- Severe pain
- Inability to bear weight
- Obvious fracture or joint deformity
- Pelvic or abdominal pain
- Signs of systemic disorder: nausea, vomiting, fever, confusion
- Visible signs of a fracture or joint deformity
- Dependent on symptoms: eye, head, ear injuries



DRUG RELATED CAUSES



Fluoroquinolones

- Associated with an increased risk of tendonitis and tendon rupture in all ages
- Rupture of the Achilles tendon has been reported most frequently
- Can occur within hours-days up until months after discontinuation
- DISCONTINUE at first sign of pain, swelling, inflammation of tendon

Statins

- Myopathy may occur (e.g. muscle pain, tenderness, stiffness, cramping, weakness or generalized fatigue)

GOALS OF THERAPY

- ✓ Relieve acute symptoms of pain and inflammation
- ✓ Reduce risk or avoid recurrences of injury
- ✓ Help the athlete return to normal weight-bearing capabilities, flexibility, strength, and range of motion
- ✓ Assist the athlete in returning to pre-injury activities safely and comfortably

NON-PHARMACOLOGIC MEASURES

ACUTE TREATMENT



P

Protection: from further injury (supportive shoes)

R

Rest the injured part for a **short** period – early mobilization aids in healing

I

Ice: wrap a cold pack in a thin damp cloth and apply to the injured area for **10 to 15 minutes four times a day for the first 48 hours or until swelling lessens**

C

Compress using an elastic bandage if there is swelling present (common in sprains)

E

Elevate: try to elevate the injured part **above the heart**

NON-PHARMACOLOGIC MEASURES

ACUTE TREATMENT

Heat therapy

- **Used 2 to 14 days post-injury**
- Local heat therapy can provide analgesia
- Joint stiffness may be limited
- Increased blood flow to the area causing vasodilation → helps to repair and remove waste inflammatory products
- Apply heat for 15 to 20 min every 2 to 4 hours as needed

PHARMACOLOGIC MEASURES

Oral or topical NSAIDs

- Decrease discomfort and swelling in patients
- **Should only be used short-term**
- The elderly are at a higher risk of experiencing side effects (e.g. dyspepsia, GI bleeding, renal impairment, increased blood pressure)
 - Weigh risks vs. benefits and consider topical agents (topical diclofenac)

Acetaminophen

- Can consider for initial treatment or ongoing musculoskeletal pain
- Less GI and cardiovascular risks compared to oral NSAIDs

Intra-articular/peritendinous corticosteroid injections

- Examples include triamcinolone acetonide, methylprednisolone acetate, and betamethasone
- Useful for treating patients with **specific injuries** (e.g. rotator cuff disorder, tennis or golfer elbow)
- Systematic review for peritendinous corticosteroid injections: short-term benefit for treating tendinopathies however effectiveness varies depending on the injury site

PHARMACOLOGIC MEASURES

Methocarbamol – available OTC and by prescription

- May be used for muscle spasms
- Not routinely recommended, can cause sedation or dizziness

Orphenadrine

- Used short-term, has many anticholinergic side effects, and causes drowsiness
- Contraindicated in bladder obstruction, angle closure glaucoma, achalasia, and myasthenia gravis

Nitroglycerin (off-label)

- Reduces pain linked with daily activities in patients experiencing chronic tendinopathies
- Side effects: **headaches**, dizziness, skin rash, hypotension (caution with anti-hypertensives)
- Contraindications: co-administration of PDE5 inhibitors and nitrates (including topical)
- Use 1/4 of a patch at a time – applied directly over affected tendon

Platelet-Rich Plasma (PRP) or autologous blood injections

- Mixed evidence – provides natural growth factors to the affected tendon
- May be an alternative to surgery

SPECIAL POPULATIONS - PREGNANCY AND BREASTFEEDING

Acetaminophen	• Safe in pregnancy and breastfeeding
NSAIDs	• Avoid, especially beyond 20 weeks gestation • Also not recommended in 1st trimester • Avoid while breastfeeding if possible
Injectable corticosteroids	• Consider if non-drug measures and acetaminophen are ineffective (conflicting evidence)

MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Symptoms e.g. pain and injury	Reduce and heal	Weeks
Patient's normal daily functioning (e.g. capable to play sports to the same capacity as before the injury)	Return to normal	Ongoing

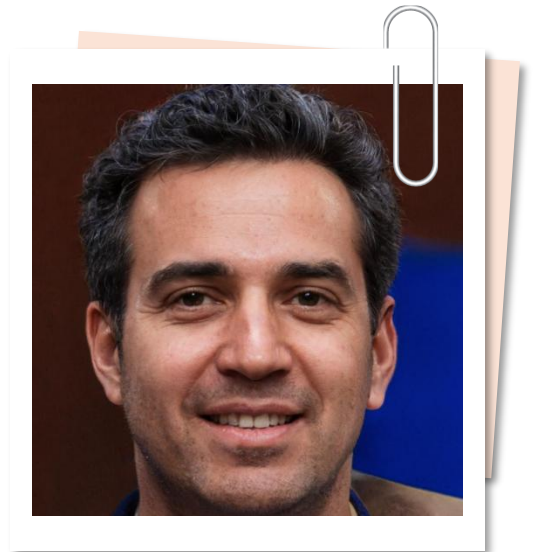
THERAPEUTIC TIPS

- Rule out red flags before making a recommendation
- Acute treatment is largely non-pharmacological: PRICE
 - Do not apply heat until at least 48 hours post-injury
- NSAIDs recommended as first-line short-term pharmacotherapy
- Consider acetaminophen for elderly patients with persistent musculoskeletal pain
- If muscle spasms are present methocarbamol or orphenadrine can be considered

CASE 3

LB is a 54-year-old male who presents with back pain that started about 3 days ago. He is seeking advice on how to help relieve it. He said he has a bottle of Tylenol® (acetaminophen) at home but has yet to try anything, because he wanted to speak to you first. He has a past medical history of hypertension, type 2 diabetes, and anxiety for which he takes amlodipine 5mg po daily, metformin SR 2000mg PO daily, lorazepam 0.5mg SL daily PRN, vitamin D 1000 IU po daily and a multivitamin PO daily. LB's hypertension and diabetes are well-controlled. He has a documented allergy to sulfa, resulting in a rash. He smokes socially and rarely drinks alcohol. He starts his day with a cup of coffee.

1. Which of the following is **NOT** a common clinical presentation of LBP?
 - a) Muscle aches
 - b) Pain radiating to the leg
 - c) Increased flexibility and ROM
 - d) Difficulty standing up straight



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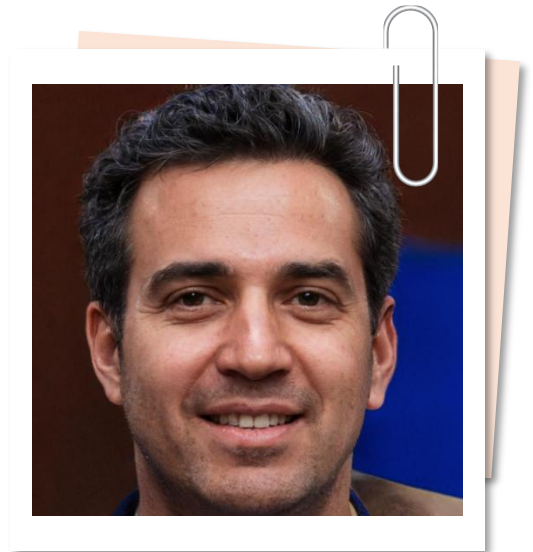
2. What is the mainstay of non-pharmacological treatment for acute, non-sciatic LBP?
- a) Encourage bedrest
 - b) Avoid bedrest
 - c) Encourage sick leave from work
 - d) Avoid activity



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3. Which of the following is an appropriate first-line option for LB to attempt treating his LBP?
- a) Tylenol 500 mg po q4-6h PRN
 - b) Oxycodone IR 5 mg po q4h PRN
 - c) Hydromorphone CR 3 mg po q12h
 - d) Naproxen 500 mg po BID



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4. LB's back pain has not yet resolved, and it is 12 weeks after onset. He is experiencing moderate signs in mood changes. Which of the following is an approved antidepressant for use in chronic back pain?
- a) Amitriptyline
 - b) Duloxetine
 - c) Venlafaxine
 - d) Escitalopram



CASE 4

SI is a 31-year-old male who comes in complaining of lower leg pain. He has recently gotten into playing basketball and a friend said he may have developed shin splints. He is seeking advice, as he cannot see his family doctor until Monday morning. He has a past medical history of anxiety and uncontrolled hypertension, and he is being treated with citalopram 20mg PO daily. In addition, he takes magnesium oxide 420 mg po at bedtime. He is a social smoker and drinks approximately 2 beers a week. He has a documented allergy to penicillin, which causes hives.

1. Which of the following is a red flag and reason for referral in assessing a sports injury?
 - a) It has been 2 days since the injury
 - b) It has been 15 days since the injury
 - c) It has been 7 days since the injury
 - d) It has been 12 hours since the injury



CASE 4

SI is a 31-year-old male who comes in complaining of lower leg pain. He has recently gotten into playing basketball and a friend said he may have developed shin splints. He is seeking advice, as he cannot see his family doctor until Monday morning. He has a past medical history of anxiety and uncontrolled hypertension, and he is being treated with citalopram 20mg PO daily. In addition, he takes magnesium oxide 420 mg po at bedtime. He is a social smoker and drinks approximately 2 beers a week. He has a documented allergy to penicillin, which causes hives.

2. Which of the following is false when it comes to the PRICE protocol?
- a) Protect the injury (wear supportive shoes)
 - b) Rest the injured area for 6 weeks; do not mobilize early
 - c) Ice for 10-15 minutes at a time at least QID for the first 48 hours
 - d) Compress with an elastic bandage if there is swelling



CASE 4

SI is a 31-year-old male who comes in complaining of lower leg pain. He has recently gotten into playing basketball and a friend said he may have developed shin splints. He is seeking advice, as he cannot see his family doctor until Monday morning. He has a past medical history of anxiety and uncontrolled hypertension, and he is being treated with citalopram 20mg PO daily. In addition, he takes magnesium oxide 420 mg po at bedtime. He is a social smoker and drinks approximately 2 beers a week. He has a documented allergy to penicillin, which causes hives.

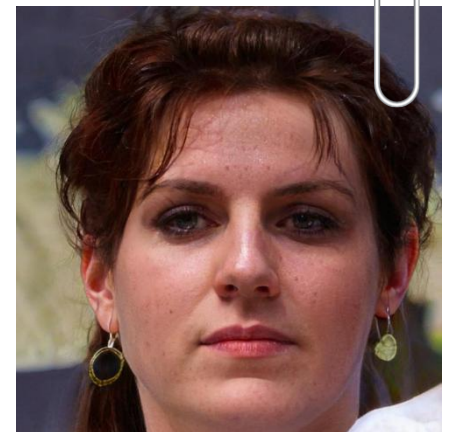
3. SI says his pain is about 2/10. What is the safest and most reasonable first-line pharmacologic option for him to trial?
- a) Suggest he get a prescription for a nitroglycerin patch from his family doctor
 - b) Suggest he get a prescription for high-dose naproxen from his family doctor
 - c) Acetaminophen 500-1000 mg po q6h prn
 - d) Ibuprofen 400 mg po q8h prn



CASE 5

LK is a 28-year-old female who comes in complaining of leg pain. She states she was trying out some new exercises recommended by her doctor and thinks she strained her leg. She is currently 16 weeks pregnant and is hoping to take a medication to help with her pain. Her obstetrician advised her to see her pharmacist for a recommendation. Her past medical history is significant for anxiety and migraines. She currently takes sertraline 50 mg daily, vitamin D 1000 units daily, ginger tablets as needed for nausea, and a prenatal vitamin daily. She does not smoke or drink alcohol but does drink a pop on occasion. She has no known allergies to medications but is allergic to the adhesive on bandages.

1. Which of the following is **CONTRAINDICATED** for LK?
 - a) Acetaminophen
 - b) Naproxen
 - c) Shockwave therapy
 - d) Injectable corticosteroids



TAKEAWAY

LOW BACK PAIN

- Acute(<4 weeks) → NSAIDs, muscle relaxants short-term
- Chronic (>3 months) → some evidence for NSAIDs, antidepressants if concomitant depression/anxiety, GABA derivatives, opioids (if moderate-severe symptoms)
- Acetaminophen is considered ineffective

NECK PAIN

- Acute neck pain (1-4 weeks) → NSAIDs, tramadol, codeine, muscle relaxants
 - Little evidence to support the use of acetaminophen
- Subacute neck pain (4-12 weeks) → IM lidocaine injection, opioids (if severe symptoms)
 - Expert opinion: duloxetine, TCAs, GABA derivatives, tramadol
 - Limited evidence: NSAIDs, muscle relaxants
- Chronic neck pain (≥ 6 months)= based on expert opinion: muscle relaxants, GABA derivatives, tramadol
 - Weak/limited evidence: NSAIDs, opioids, epidural corticosteroids, and lidocaine
 - Not effective: TCAs, botulinum toxin A

SPORTS INJURIES

- Acute sports injuries → PRICE +/- pharmacologic symptomatic treatment
- NSAIDS (short-term use only), muscle relaxants (for muscle spasms – short-term use only)

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CHANGE LOG

August 2025

- Formatting changes made
- Slides 10, 25 and 26: Re-worded and rephrased information

November 2025

- Minor formatting edits, no content changes made

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 10: Minor updates to red flags, minor phrasing and formatting
- Slide 36: Minor rephrasing to case. Question and answer choices remain unchanged
- Slide 42: Removed yellow flags and updated red flags
- Slide 49: Updated recommendations on NSAIDs